Start Minikube

First start the cluster normally (this is usually already done in the lab).

Open **Minikube Dashboard**:

```
minikube dashboard
```

A browser UI will open.

You will see the **Kubernetes Dashboard interface**.

Open Kubernetes Dashboard

Dashboard sections look like:

- Workloads
- Services
- Config
- Storage
- Discovery
- **Horizontal Pod Autoscaler**
- Nodes

This UI lets you create everything **without commands**.

Create Deployment (AI Application)

Step-by-Step

1. Go to **Workloads → Deployments**
2. Click **Create**
3. Select **Create from form**

Fill the fields:

App Name

```
Ai-app
```

Container Image

```
Ai-model
```

Number of Pods

```
2
```

Container Port

```
5000
```

Click **Deploy**

The dashboard will create:

- Deployment
- Pods

You will see **2 running pods**.

Check Pods

Go to:

```
Workloads → Pods
```

You will see:

```
ai-app-xxxx
ai-app-yyyy
```

Status should be **Running**.

Create Service (Expose Application)

Go to:

```
Discovery and Load Balancing → Services
```

Click **Create**

Choose **Create from form**

Fill:

Service Name

```
Ai-service
```

Selector

```
app: ai-app
```

Port

```
5000
```

Target Port

```
5000
```

Service Type

```
NodePort
```

Click **Create**

Now the service will appear.

Access Application

Open the service.

You will see:

```
NodePort
ClusterIP
Port
```

Open the **NodePort URL**.

Output will show:

```
AI Model Running in Kubernetes
```

Your application is now **running inside Kubernetes**.

Enable Metrics Monitoring

In the dashboard go to:

```
Workloads → Pods
```

You will see **CPU and Memory usage columns**.

Example:

```
CPU: 2m
Memory: 20Mi
```

This means monitoring is active.

Create Horizontal Pod Autoscaler (UI Method)

Go to:

```
Workloads → Horizontal Pod Autoscalers
```

Click **Create**

Fill:

Name

```
Ai-app-hpa
```

Target Deployment

```
Ai-app
```

Minimum Pods

```
2
```

Maximum Pods

```
10
```

Target CPU Utilization

```
50%
```

Click **Create**

Now autoscaling is enabled.

Monitor Scaling

Go to:

```
Workloads → Horizontal Pod Autoscalers
```

You will see:

```
Current CPU
Target CPU
Replicas
```

Example:

```
CPU: 10% / 50%
Replicas: 2
```

Observe Pods

Go to:

```
Workloads → Pods
```

When load increases you will see:

```
ai-app-1
ai-app-2
ai-app-3
ai-app-4
```

Pods increase automatically.

Monitor Health

Click any pod.

You can see:

- CPU usage
- Memory usage
- Logs
- Events
- Container status

This is **health monitoring**.

Monitor Performance Metrics

Go to:

```
Nodes → minikube
```

You will see:

- CPU usage
- Memory usage
- Pod distribution

This shows **cluster performance**.

Final Architecture

```
User Request
      ↓
Service (NodePort)
      ↓
Deployment
      ↓
Pods (AI App)
      ↓
Metrics Server
      ↓
Horizontal Pod Autoscaler
```

What You Demonstrated

Your assignment now shows:

AI app deployed in **Kubernetes**
Service exposing application
CPU and memory monitoring
Auto-scaling using HPA
Health monitoring using dashboard